SECTION A - STRUCTURED QUESTIONS

INSTRUCTIONS: Answer ALL of the following questions. Write your answers in the answer booklet.

Question 1 (15 marks)

a) Software design fundamentals establish the foundation for creating reliable, maintainable, and scalable software systems that meet the needs of users and stakeholders while being adaptable to changing requirements and technological landscapes. Describe **TWO(2)** software design fundamentals based on your learning.

(6 Marks)

b) Define **THREE (3)** states and provide an example of each state during engineering problem-solving.

(9 Marks)

Question 2 (20 marks)

a) Provide **THREE(3)** major software engineering design challenges.

(6 Marks)

b) Name **TWO(2)** UML structural diagrams and **TWO(2)** behavioral diagrams.

(4 Marks)

c) Given the **"Make Invoice"** use case description (Refer to Table 1 below) for a Fleet Management System, draw a complete Sequence Diagram.

(10 Marks)

Table 1 Make Invoice Use case Specifications

Use Case Name	Make Invoice
Description	The system generates an invoice for the services provided in the Fleet Management System, detailing costs associated with vehicle usage, maintenance, and additional charges.
Actors	Operator, Billing System
Preconditions	Operator has logged into the Fleet Management System. Relevant data for billing is available and up-to-date.
Basic Flow	1. Operator selects the "Make Invoice" option from the system interface. 2. System prompts the operator to choose a specific client or project for which the invoice will be generated. 3. Operator selects the client or project. 4. System calculates the costs based on vehicle usage, maintenance records, and any additional charges. 5. System generates and displays the invoice. 6. Operator reviews the invoice. 7. Operator confirms the invoice. 8. System stores the invoice data for future reference. 9. Billing System is notified to process the invoice.
Alternate Paths	4a. If there are no additional charges or maintenance records, skip the relevant calculation steps. 6a. If the operator identifies an error in the invoice, they can choose to edit the invoice.
Postconditions	The generated invoice data is stored in the system for future reference. The Billing System has been notified to process the generated invoice.
Extension Point	**<Calculate Discounts>** **<Invoice Generation for Late Fees>**

Question 3 **(14 marks)**

a) There are several key tasks that need to be performed during the requirement elicitation stage. Explain **FOUR (4)** of the key tasks that need to be performed during the requirements elicitation stage.

(8 marks)

b) Name and provide descriptions of **TWO(2)** quality attributes of software systems.

(6 marks)

Question 4 **(11 marks)**

a) Explain the importance of the Logical View in the 4+1 Architectural View Model. Provide examples of elements that might be represented in the Logical View and discuss their significance in the overall system design.

(5 marks)

b) Specify **THREE(3)** quality attributes that can be evaluated during the Process View in the 4+1 Architectural View Model.

(6 marks)

SECTION B - ESSAY / PROBLEM SOLVING

INSTRUCTIONS: Answer ALL of the following questions. Write your answers in the answer booklet.

Question 1 **(18 marks)**

You are overseeing the development of an innovative online collaboration platform utilizing a client-server architecture. This platform aims to enhance real-time communication and document sharing among users.

a) Provide **TWO(2)** strengths of the client-server architecture and provide examples to illustrate each point. Highlight how these strengths contribute to the overall software qualities of client-server systems.

(4 marks)

b) Provide **TWO(2)** weaknesses inherent in the client-server architecture and discuss potential challenges that developers may encounter. Provide recommendations on how these weaknesses can be mitigated or minimized in the design and implementation of client-server systems.

(6 marks)

c) Draw a Sequence Diagram (SD) illustrating the process of a user uploading a document to the collaboration platform. Highlight the interactions between the client and server components. Provide annotations to explain the sequence of events and the data exchanged during this process.

(8 marks)

Question 2 **(22 marks)**

Case Study: Fleet Management System

You are the lead software engineer at a technology company, and you are tasked with developing a Fleet Management System that manages different types of vehicles. The system should be extensible to accommodate various kinds of vehicles, such as cars, bicycles, and electric scooters. The goal is to ensure that the system adheres to the Liskov Substitution Principle (LSP) and Open-Closed Principle (OCP).

a) Define the Liskov Substitution Principle and explain how it promotes code reusability and extensibility in object-oriented programming. Provide examples to illustrate the concept.

(4 marks)

b) Design a class diagram for the vehicle management system, incorporating the Liskov Substitution Principle. Consider different types of vehicles and their common behaviors.

(5 marks)

c) Consider a scenario where a new type of vehicle, a motorcycle, needs to be added to the system. Based on the answer on Question 2(b), redraw the whole class diagram of how the Open-Closed Principle can be applied to extend the functionality of the vehicle management system.

(5 marks)

d) Based on the answer in Question 2(c), write a code snippet in a programming language of your choice that demonstrates both Liskov Substitution Principle and Open-Closed Principle within the vehicle management system.

(8 marks)

- QUESTIONS END -